西湖大學
WESTLAKE UNIVERSITY

Bioinformatics training — 3

RNA-seq数据分析

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Dec 16th, 2021

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- 利用RNA-seq 检测差异表达基因

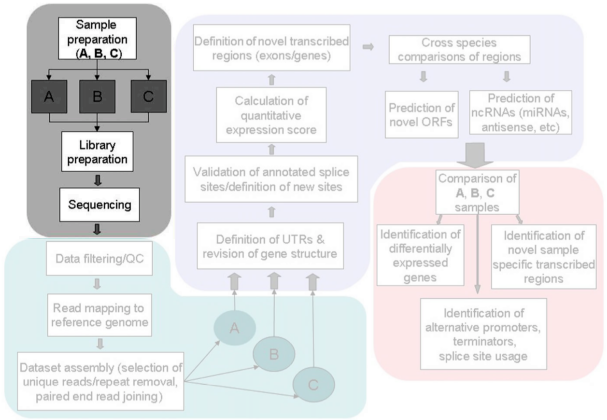
转录组

- 转录组：是指在特定的发育阶段和生理条件下细胞中转录产生的所有转录物，包括mRNA、rRNA、tRNA和其他的非编码RNA，狭义上也可以指细胞所能转录出的所有mRNA，这个术语适用的范围是一个给定的有机体、组织或者特定的细胞集合。
- 转录组学：是一门从整体水平上研究基因转录情况和转录调控规律的学科。转录组学研究分为序列水平与表达量水平两个层面的内容，这两个层面综合在一起完整的反映了生物体遗传信息的表达方式。
  - 序列水平的研究：包括对未知转录本的探索以及对同一个基因在不同样品中的序列差异分析，包括起始密码子鉴定、内含子边界确定、UTR确定、可变剪切、单核苷酸多态性位点研究。
  - 表达量水平的研究：可以加深了解生物体基因调节的过程和特殊性状产生的分子机制，包括多种 RNA表达量研究、新转录本的检测等。

RNA-seq

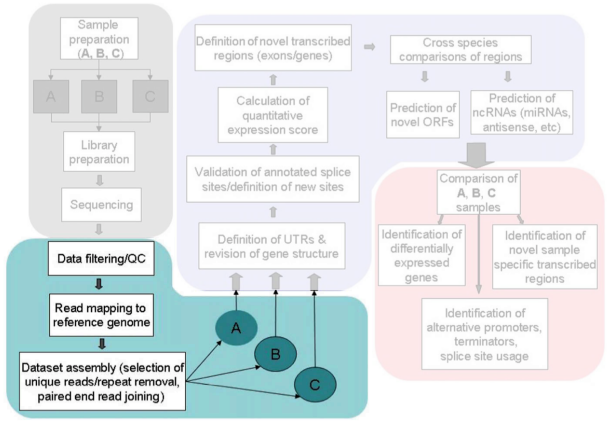
- 把高通量测序技术应用到由 RNA 逆转录生成的 cDNA 上，从而获得来自不同基因的RNA 片段在特定样本中的含量，这就是 RNA测序（RNA-seq）。

第一步：通过湿实验进行信息提取。包括样本的制备、文库构建和测序。

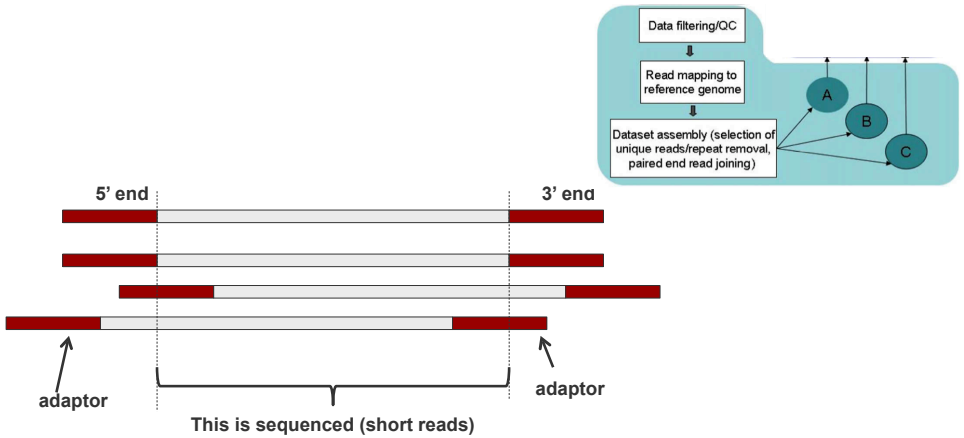


{Wilhelm, B.T., *et al.*, *Methods*, 2009}

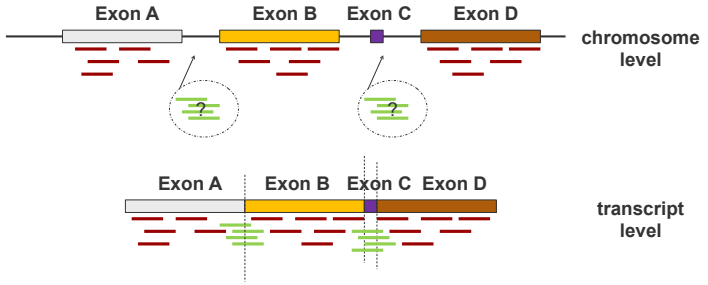
第二步：数据预处理。包括对测序数据的质量控制、测序片段的筛选和回帖。



{Wilhelm, B.T., *et al.*, *Methods*, 2009}



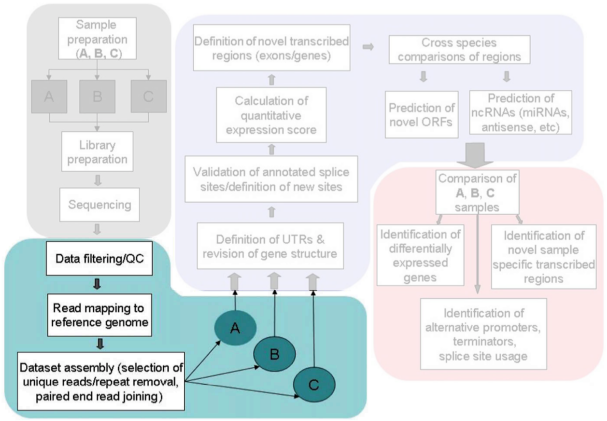
Adaptor and reads in RNA-seq



Reads' mappings at chromosome and transcript level

RNA-seq研究策略

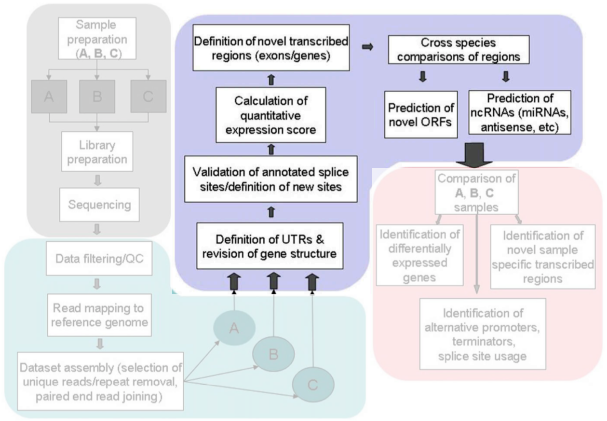
第二步：数据预处理。包括对测序数据的质量控制、测序片段的筛选和回帖。



{Wilhelm, B.T., *et al.*, *Methods*, 2009}

RNA-seq研究策略

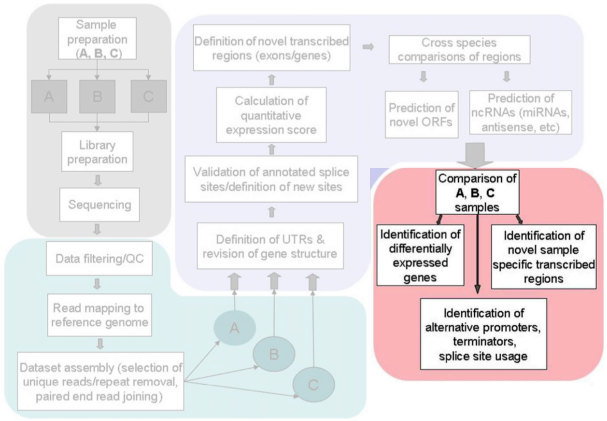
第三步：样品特异性数据分析。针对样品特征如不同组织、发育阶段、生长条件和待解决生物学问题进行特异性的数据分析。



{Wilhelm, B.T., *et al.*, *Methods*, 2009}

RNA-seq研究策略

第四步：交叉样品分析。即比较不同实验条件下的样本的转录组差异，包括差异表达分析、样本特异性转录本分析、可变转录元件分析等。



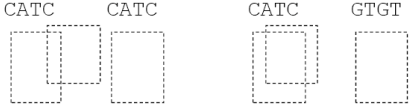
{Wilhelm, B.T., *et al.*, *Methods*, 2009}

序列回帖



A DNA Sequence: ACATAGGATCATGAAGTACCCATATCTAGTGGG
reads: AGGA, CATC, ATAT, TTTG, GTGT

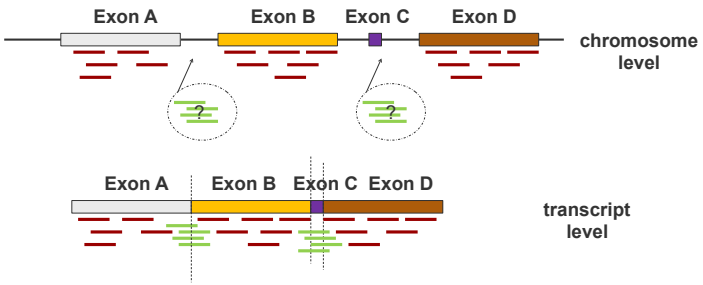
Matched Results: ACATAGGATCATGAAGTACCCATATCTAGTGGG
Perfect Match: AGGA ATAT
1bp Mismatch: CATC CATC CATC GTGT





- 在参考基因组上找到片段序列的匹配位置。
- 效率很关键，因为一次高通量测序包含有至少几百万个测序片段。

明确序列回帖的目的



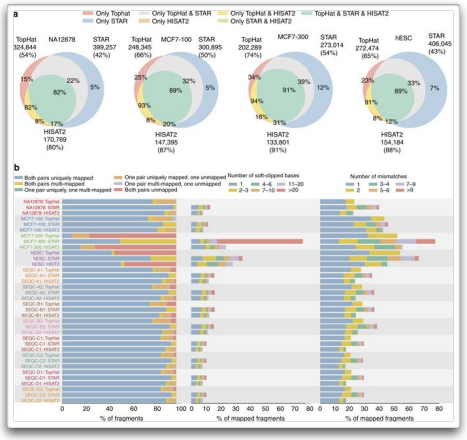
确定表达量?	寻找新的转录本? 检测 RNA可变剪切?
Bowtie、BWA	TopHat、HISAT2、STAR

常用的短片段序列回帖工具

Software	Max. mismatches	Gap	Max. read length	Report All matches	Reference
ELAND	2	No	32	N	A. Cox (Illumina)
SOAP	2	1	60	N	R. Li (2008)
RMAP	>4	No	64	N	A. D. Smith (2008)
SeqMap	>4	>4	>200	Y	H. Jiang (2008)
ZOOM!	>4	1	64	Y	H. Lin (2008)
MAQ	3	No	64	N	H. Li (2008)
Bowtie	3	No	>200	Y	B. Langmead (2009)
STAR	—	—	>200	Y	A. Dobin (2013)
HISAT	—	—	>200	Y	D. Kim (2015)

- https://wiki2.org/en/List_of_sequence_alignment_software#Short-read_sequence_alignment

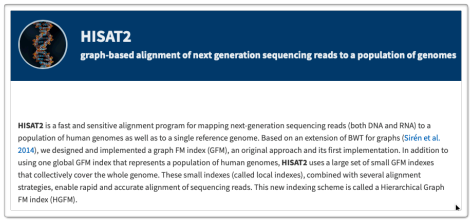
如何选取序列回帖工具



{Sahraeian, S.M.E, *et al., Nat Communications*, 2017}

- HISAT2找到的可变剪切位点正确率最高，但是在总数上比TopHat和STAR少。即HISAT2的II型错误(假阴性）较少，但是I型错误（假阳性）较多。
- STAR的稳定性也是最佳。
- HISAT2比STAR和TopHat平均快上2.5~100倍

利用 HISAT2进行序列回帖



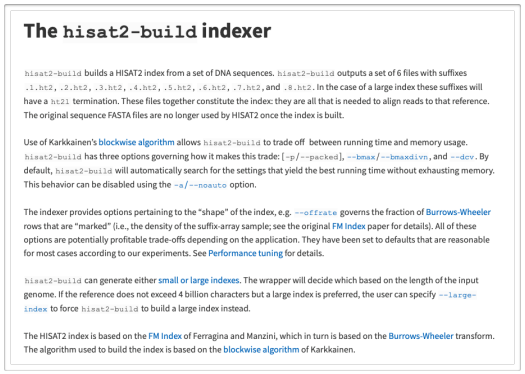
{Kim, D, *et al., Nat Methods*, 2015}

- HISAT是一个利用大量覆盖整个基因组的FM索引，从而能够将 RNA-seq 测序片段与基因组进行快速比对的序列回帖工具。

HISAT2—参考基因组索引

- 构建





HISAT2—比对命令

```
zhaochengchen@BioServer1:~$ hisat2 --help
HISAT2 version 2.1.0 by Daehwan Kim (infphilo@gmail.com, www.ccb.jhu.edu/people/infphilo)
Usage:
  hisat2 [options]* -x <ht2-idx> {-1 <m1> -2 <m2> | -U <r> | --sra-acc <SRA accession number>} [-S <sam>]

  <ht2-idx>  Index filename prefix (minus trailing .X.ht2).
  <m1>       Files with #1 mates, paired with files in <m2>.
             Could be gzip'ed (extension: .gz) or bzip2'ed (extension: .bz2).
  <m2>       Files with #2 mates, paired with files in <m1>.
             Could be gzip'ed (extension: .gz) or bzip2'ed (extension: .bz2).
  <r>        Files with unpaired reads.
             Could be gzip'ed (extension: .gz) or bzip2'ed (extension: .bz2).
  <SRA accession number>      Comma-separated list of SRA accession numbers, e.g. --sra-acc SRR353653,SRR353654.
  <sam>      File for SAM output (default: stdout)

  <m1>, <m2>, <r> can be comma-separated lists (no whitespace) and can be
  specified many times.  E.g. '-U file1.fq,file2.fq -U file3.fq'.
```

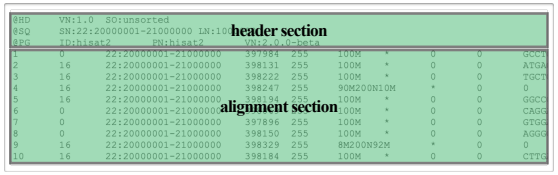
- 主要参数：
  - -x <hisat2-idx>
    - 参考基因组索引文件的前缀。
  - -1 <m1>
    - 双端测序结果的第一个文件。若有多组数据，使用逗号将文件分隔。Reads的长度可以不一致。
  - -2 <m2>
    - 双端测序结果的第二个文件。若有多组数据，使用逗号将文件分隔，并且文件顺序要和-1参数对应。
  - -U <r>
    - 单端数据文件。若有多组数据，使用逗号将文件分隔。可以和-1、-2参数同时使用。Reads的长度可以不一致。
  - -S <hit>
    - 指定输出SAM格式文件。
  - -o
    - 指定输出文件的名字。

HISAT2—比对结果

- 回帖情况的统计。

```
20000 reads; of these:
  20000 (100.00%) were unpaired; of these:
    1247 (6.24%) aligned 0 times
    18739 (93.69%) aligned exactly 1 time
    14 (0.07%) aligned >1 times
93.77% overall alignment rate
```

- 以 SAM 格式输出的回帖结果。

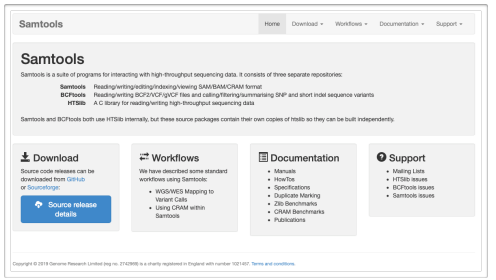


Col	Field	Type	Regexp/Range	Brief description
1	QNAME	String	[!-?A-~]{1,254}	Query template NAME
2	FLAG	Int	$[0, 2^{16}-1]$	bitwise FLAG
3	RNAME	String	*|[:rname:^*=][:rname:]*	Reference sequence NAME[11]
4	POS	Int	$[0, 2^{31}-1]$	1-based leftmost mapping POSition
5	MAPQ	Int	$[0, 2^{8}-1]$	MAPping Quality
6	CIGAR	String	*|([0-9]+[MIDNSHPX=])+	CIGAR string
7	RNEXT	String	*|=|[:rname:^*=][:rname:]*	Reference name of the mate/next read
8	PNEXT	Int	$[0, 2^{31}-1]$	Position of the mate/next read
9	TLEN	Int	$[-2^{31}+1, 2^{31}-1]$	observed Template LENgth
10	SEQ	String	*|[A-Za-z=.]+	segment SEQuence
11	QUAL	String	[!-~]+	ASCII of Phred-scaled base QUALity+33

HISAT2—结果保存

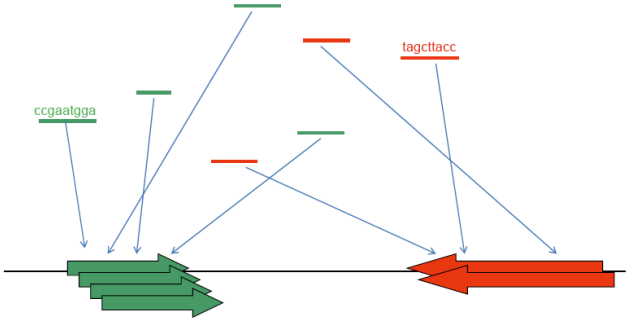
- 用Samtools将SAM文件转为BAM文件，并且排序索引好。

```
samtools view -S your_file_name.sam -b > your_file_name.bam
samtools sort your_file_name.bam -o your_file_name.sorted.bam
samtools index your_file_name.sorted.bam
```



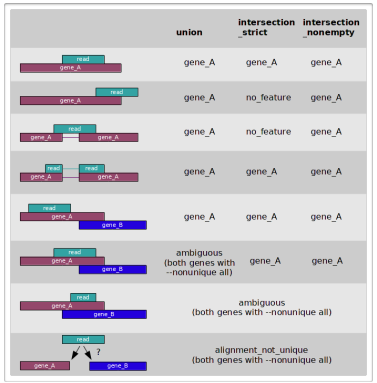


基因表达谱的获取



- 基因的表达水平决定 <-> 落在该基因上的reads个数。

利用htseq-count统计基因上片段数目



- 如何判断一个 reads 属于某个基因，htseq-count 提供了 union，intersection_strict，intersection_nonempty 3 种模型。

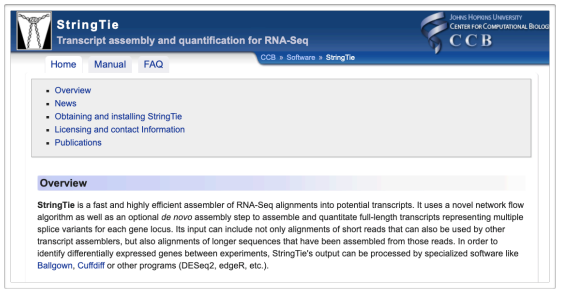
基因表达水平标准化

- 如果基因的转录区域越长，那么会有越多的reads被回帖到基因上；
- 如果测序的深度越深，那么会有越多的reads被回帖到基因上。

FPKM/RPKM	TPM
Fragments/Reads per Kilobase Million $FPKM/RPKM = \frac{Number\ of\ fragments/reads\ mapped\ to\ genes}{Size\ of\ transcript\ in\ kb \times Number\ of\ total\ mapped\ reads(M)}$	Transcripts Per Million $Adjust\ counts = \frac{Number\ of\ fragments/reads\ mapped\ to\ genes}{Size\ of\ transcript\ in\ kb}$ $TPM = \frac{Adjust\ counts\ of\ fragments/reads\ mapped\ to\ genes}{Adjust\ counts\ of\ total\ mapped\ reads(M)}$

利用 StringTie获得基因表达值

- StringTie应用流神经网络算法和可选的de novo组装进行转录本组装并预计表达水平。StringTie实现了更完整、更准确的基因重建，并更好地预测了表达水平。



StringTie的下载与安装

Obtaining and installing StringTie

The current version of StringTie can be downloaded as precompiled binary or as a source package:

- stringtie-2.1.1.tar.gz : source package
- stringtie-2.1.1.Linux_x86_64.tar.gz : Linux x86_64 binary package
- stringtie-2.1.1.OSX_x86_64.tar.gz : Apple OS X binary package (for OS X v10.7 and above)

In order to build and install StringTie from the source package the following steps can be taken:

1. Unpack the downloaded StringTie source archive in a directory of your choice, e.g.:

```
cd ~/src/
tar xvfz ~/Downloads/stringtie-VER.tar.gz
```

   A directory called `stringtie-VER` (where VER is the current numeric version of the program) will be created in the current directory.

2. Change directory and build the `stringtie` executable:

```
cd stringtie-VER
make release
```

   Alternatively, the source tree can be downloaded from GitHub and built in a similar fashion:

```
git clone https://github.com/gpertea/stringtie
cd stringtie
make release
```

3. Optionally, the `stringtie` executable can be copied to one of the shell's PATH directories for easy access.

- 下载对应系统的源代码安装压缩包。
- 解压并安装。

StringTie的运行

Running StringTie

Run `stringtie` from the command line like this:

```
stringtie <aligned_reads.bam> [options]*
```

The main input of the program is a BAM file with RNA-Seq read mappings which must be sorted by their genomic location (for example the `accepted_hits.bam` file produced by TopHat or the output of HISAT2 after sorting and converting it using samtools as explained below).

Input files

StringTie takes as input a binary SAM (BAM) file sorted by reference position. This file contains spliced read alignments and can be produced directly by programs such as TopHat or it can be obtained by converting and sorting the output of HISAT2. We recommend using HISAT2 as it is a fast and accurate alignment program. A text file in SAM format which was produced by HISAT2 must be sorted and converted to BAM format using the samtools program:

```
samtools view -Su alns.sam | samtools sort - alns.sorted
```

The file resulted from the above command (`alns.sorted.bam`) can be used as input for StringTie.

Every spliced read alignment (i.e. an alignment across at least one junction) in the input SAM file must contain the tag `XS` to indicate the genomic strand that produced the RNA from which the read was sequenced. Alignments produced by TopHat and HISAT2 (when run with --dta option) already include this tag, but if you use a different read mapper you should check that this XS tag is included for spliced alignments.

NOTE: be sure to run HISAT2 with the --dta option for alignment, or your results will suffer.

As an option, a reference annotation file in GTF/GFF3 format can be provided to StringTie. In this case, StringTie will prefer to use these "known" genes from the annotation file, and for the ones that are expressed it will compute coverage, TPM and FPKM values. It will also produce additional transcripts to account for RNA-seq data that aren't covered by (or explained by) the annotation. Note that if option -e is not used the reference transcripts need to be fully covered by reads in order to be included in StringTie's output. In that case, other transcripts assembled from the data by StringTie and not present in the reference file will be printed as well.

NOTE: we highly recommend that you provide annotation if you are analyzing a genome that is well-annotated, such as human, mouse, or other model organisms.

- 该输入文件要求必须按其基因组位置排序。

StringTie的结果

Output files

Main Outputs

1. Stringtie's main output is a GTF file containing the assembled transcripts
2. Gene abundances in tab-delimited format
3. Fully covered transcripts that match the reference annotation, in GTF format
4. Files (tables) required as input to Ballgown, which uses them to estimate differential expression
5. In merge mode, a merged GTF file from a set of GTF files

1. StringTie's primary GTF output

The primary output of StringTie is a Gene Transfer Format (GTF) file that contains details of the transcripts that StringTie assembles from RNA-Seq data. GTF is an extension of GFF (Gene Finding Format, also called General Feature Format), and is very similar to GFF2 and GFF3. The field definitions for the 9 columns of GTF output can be found at the Ensembl site here. The following is an example of a transcript assembled by StringTie as shown in a GTF file (scroll right within the box to see the full field contents):

```
seqname source      feature     start   end     score   strand  frame attributes
chrX   StringTie   transcript  281394  303355  1000    +       .     gene_id "ERR18
chrX   StringTie   exon        281394  281684  1000    +       .     gene_id "ERR18
...
```

2. Gene abundances in tab-delimited format

If StringTie is run with the `-A <gene_abund.tab>` option, it returns a file containing gene abundances. The tab-delimited gene abundances output file has nine fields; here is an example of a gene abundance file produced by StringTie using reference annotation:

```
Gene ID     Gene Name   Reference   Strand  Start   End     Coverage    FPKM      T
NM_000451   SHOX        chrX        +       624344  646823  0.000000    0.000000  0
NM_006883   SHOX        chrX        +       624344  659411  0.000000    0.000000  0
...
```

- 输出文件：

1、 GTF文件： 记录组装的转录本信息

2、 Tab文件： 记录基因丰度信息

StringTie的结果

3. Fully covered transcripts matching the reference annotation transcripts (in GTF format)

If StringTie is run with the `-C <cov_refs.gtf>` option (requires `-G <reference_annotation>`), it returns a file with all the transcripts in the reference annotation that are fully covered, end to end, by reads. The output format is a GTF file as described above. Each line of the GTF is corresponds to a gene or transcript in the reference annotation.

4. Ballgown Input Table Files

If StringTie is run with the `-B` option, it returns a Ballgown input table file, which contains coverage data for all transcripts. The output table files are placed in the same directory as the main GTF output. These tables have these specific names: (1) e2t.ctab, (2) e_data.ctab, (3) i2t.ctab, (4) i_data.ctab, and (5) t_data.ctab. A detailed description of each of these five required inputs to Ballgown can be found on the Ballgown site, at this link.

5. Merge mode: Merged GTF

If StringTie is run with the `--merge` option, it takes as input a list of GTF/GFF files and merges/assembles these transcripts into a non-redundant set of transcripts. This step creates a uniform set of transcripts for all samples to facilitate the downstream calculation of differentially expressed levels for all transcripts among the different experimental conditions. Output is a merged GTF file with all merged gene models, but without any numeric results on coverage, FPKM, and TPM. Then, with this merged GTF, StringTie can re-estimate abundances by running it again with the `-e` option on the original set of alignment files, as illustrated in the figure below.

- 输出文件：

3、 GTF文件：完全覆盖与参考注释基因组文件所匹配的转录本信息

4、 *.ctab文件：用于Ballgown软件做差异表达分析的输入文件

5、 GTF文件：在合并模式下，生成一个合并的GTF文件

Run RNA-seq analysis pipeline in Peilab server

```
RunRNAseq /datapath \
          /resultpath \
          sample_name \
          pair \
          YourData_r1.fq.gz,YourData_r2.fq.gz \
          /data/zhaochengchen/Data/mm9/mm9 \
          /data/zhaochengchen/Data/mm9/mm9_refSeq.gtf \
          /data/zhaochengchen/Data/mm9/mm9.refseq.bed \
          /data/zhaochengchen/Data/mm9/mm9.len
```

datapath	The path where your data is
resultpath	The path where the result is
sample_name	Sample name
layout	Paired end or single end
fastq_files	Name of fastq files
mapping_index	Reference index you want to map
annotation in gtf format	The path where gene annotation file (.gtf file) is
annotation in bed format	The path where gene annotation file (.bed file) is
chromosome information	The path where .len file is

RNA-seq 数据的下游分析

- RNA-seq数据的下游分析与芯片数据的下游分析目的类似，都主要是为了鉴定某些基因表达变化的显著性或对基因表达模式进行识别。常用的方法包括监测两个或多个样品基因表达水平比值的表达差异性，减少维数并进行归类的主成分分析，以及用做类型发现和类型预测的聚类分析。
- RNA-seq的测序片段数目普遍认为符合泊松分布。

检测样本间的差异表达基因

基因表达值

	Treatment.rep1	Treatment.rep2	Treatment.rep3	Control.rep1	Control.rep2	Control.rep3
gene.1	7618	5276	838	10	48	51
gene.2	32	71	361	3470	8266	7958
gene.3	623	7371	202	218	924	495
gene.4	4266	4007	2558	0	732	823
...	...	...	...	...	...	
...	...	...	...	...	...	
...	...	...	...	...	...	
gene.n	926	1440	1876	740	588	262

样本信息

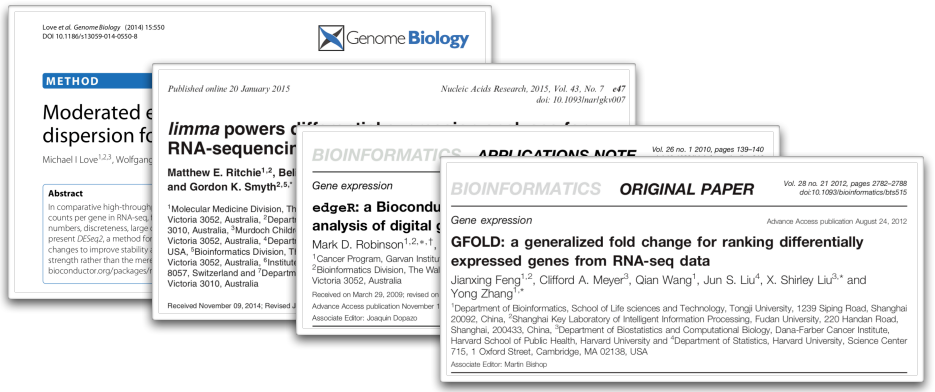
SampleName	Treatment/Control
Treatment.rep1	Treatment
Treatment.rep2	Treatment
Treatment.rep3	Treatment
Control.rep1	Control
Control.rep2	Control
Control.rep3	Control

↓?

差异表达基因

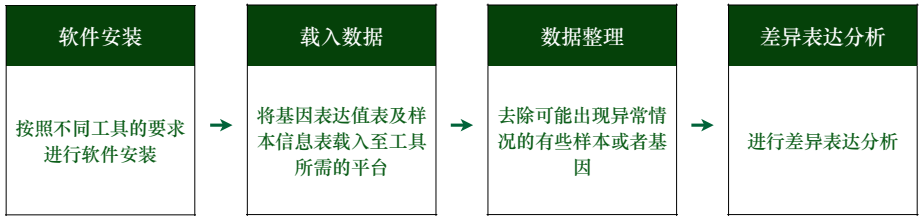
gene.1	gene.11	gene.199	gene.299	...	gene.n

常见的几种差异表达分析工具



DESeq2	limma	edgeR	GFOLD

差异表达分析流程



几种差异表达分析工具的安装

软件安装

DESeq2

```r
# start R (version "3.6") and enter
install.packages("BiocManager")
BiocManager::install("DESeq2")
```

limma

```r
# start R (version "3.6") and enter
install.packages("BiocManager")
BiocManager::install("limma")
```

edgeR

```r
# start R (version "3.6") and enter
install.packages("BiocManager")
BiocManager::install("edgeR")
```

GFOLD

```
# before install GFOLD,gsl has to be installed
# Then download the zip file of GFOLD
wget https://zhanglab.tongji.edu.cn/softwares/GFOLD/gfold.V1.1.4.tar.gz
make
g++ -O3 -Wall -g main.cc -o gfold -lgsl -lgslcblas \
    -I /your/installed/path/include -L /your/installed/path/lib
```


载入数据

载入数据

DESeq2

```r
# R
library("DESeq2")
expr <- read.table("mRNAseq_ExpressionMatrix.txt",sep = '\t',row.names=1,header=T)
```

limma

```r
# R
library("limma")
expr <- read.table("mRNAseq_ExpressionMatrix.txt",sep = '\t',row.names=1,header=T)
```

edgeR

```r
# R
library("edgeR")
expr <- read.table("mRNAseq_ExpressionMatrix.txt",sep = '\t',row.names=1,header=T)
```

GFOLD

```
# shell
gfold count -ann reference.gtf -tag sample1.sam -o sample1.read_cnt
gfold count -ann reference.gtf -tag sample2.sam -o sample2.read_cnt
```


数据整理

数据整理

DESeq2 limma edgeR

```r
# 去除低表达的基因
expr = expr[rowMeans(expr)>1,]
# 表达矩阵分组(癌症组织和癌旁组织)
treatment <- c("Treatment_rep1","Treatment_rep2","Treatment_rep3")
control <- c("Control_rep1","Control_rep2","Control_rep3")
treatment_expr <- expr[,treatment]
control_expr <- expr[,control]
group_list <- c(rep('treatment',ncol(treatment_expr)),rep('control',ncol(control_expr)))
```

差异表达分析

DESeq2

limma

edgeR

GFOLD

```
# shell
gfold diff -s1 sample1 -s2 sample2 -suf .read_cnt -o sample1VSsample2.diff
```


差异表达分析

DESeq2

limma

edgeR

GFOLD

```
# shell
gfold diff -s1 sample1 -s2 sample2 -suf .read_cnt -o sample1VSsample2.diff
```


差异表达分析

DESeq2

limma

edgeR

GFOLD

```
# shell
gfold diff -s1 sample1 -s2 sample2 -suf .read_cnt -o sample1VSsample2.diff
```


差异表达分析

DESeq2

limma

edgeR

GFOLD

```
# shell
gfold diff -s1 sample1 -s2 sample2 -suf .read_cnt -o sample1VSsample2.diff
```

不同方法的比较

- DESeq2、limma、edgeR和 GFOLD大部分情况下检测差异表达基因的效果没有特别明显的差异。
- GFOLD尤其适用于某些没有重复样本时的差异表达分析。

Exercise 5

- 下载GSE89056中hiF-T时期和niPSC时期的RNA-seq数据。
- 使用fastQC进行数据质量排查。
- 获得每套数据的表达谱。
- 使用至少两种方法获得差异表达基因。
- 请计算不同方法获得差异表达基因中共有的基因个数。